

JavaScript

The Scripting Language of the Web

JavaScript is used in million of web pages to add functionallity, validate forms, detect browsers, and much more.

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✓ **Client Side Scripting**

- Executed @ Client browser
- He can simply view the source code
- Examples: JavaScript, VBScript

✓ **Server Side Scripting**

- Executed @ the server
 - Return Back Client Side like (HTML, CSS, JS) that browser understand
 - User can't see the Source code
 - Use only can view the output (HTML, CSS, JS)
 - Examples: PHP, ASP, JSP
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Programming VS Scripting

- Script is interpreted line by line every time it runs
 - Program is compiled one time to be converted from readable form to machine language (binary code) which is delivered to user as a program (.exe file for example)
 - Scripting language is easy 2 learn
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JavaScript:

- ✓ Most popular Scripting Language
 - ✓ Work in major browsers like Firefox, Opera, Safari, IE
 - ✓ You should have basics of HTML/XHTML
 - ✓ It is designed to add interactivity to HTML Page
 - ✓ It is interpreted language (there is no compilation)
 - ✓ It free ☺ and client side ;)
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Java VS JavaScript

- Java (By SUN) a powerful programming language like C, C++, C#
 - It used to make desktop applications
 - Also used in server side scripting (JSP)
 - JavaScript is a Client Side Scripting Language (lightweight programming language)
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Note:

- JavaScript's Official name is ECMAScript
- Developed and maintained by **ECMA Organization**
- It was invented by Brendan Eich at Netscape (With navigator 2.0)

- You can write JavaScript in Head or Body

```
<html>
<body>
<script type="text/javascript">
document.write("Hello World!");
</script>
</body>
</html>
```

- You can make external Javascript file and call it from HTML
- Create (file.js)

```
<html>
<head>
<script type="text/javascript" src="file.js"></script>
</head>
<body>
</body>
</html>
```

- JavaScript is case Sensitive
- To make comment (in file.js) write this

```
// this is single-line comment

/*
This is multi-line comment
This is multi-line comment
*/
```

Tips:

- Use comments to prevent excusion
- Use comments at the end of the line

Note:

- JavaScript in Body section will be excuted while the page loads
- JavaScript in Head Section will be excuted when called

Variables

- Case sensitive
- Names must begin with letter or underscore

```
var x; // declare variable

x = 5; // assign value to variable

// declare & assign
var name="mohamed";

// you can also ignore "var" and it will be auto-declared

name="ahmed";

document.write(name);
```

Operators

➤ Arithmetic operators

Given that $y=5$, the table below explains the arithmetic operators:

Operator	Description	Example	Result
+	Addition	$x=y+2$	$x=7$
-	Subtraction	$x=y-2$	$x=3$
*	Multiplication	$x=y*2$	$x=10$
/	Division	$x=y/2$	$x=2.5$
%	Modulus (division remainder)	$x=y\%2$	$x=1$
++	Increment	$x=++y$	$x=6$
--	Decrement	$x=--y$	$x=4$

➤ Assignment Operators

Given that $x=10$ and $y=5$, the table below explains the assignment operators:

Operator	Example	Same As	Result
=	$x=y$		$x=5$
+=	$x+=y$	$x=x+y$	$x=15$
-=	$x-=y$	$x=x-y$	$x=5$
=	$x=y$	$x=x*y$	$x=50$
/=	$x/=y$	$x=x/y$	$x=2$
%=	$x\%=y$	$x=x\%y$	$x=0$

- + operator: used to add string var or text together

```
x = "Computer";

y = "Systems" ;

document.write(x+" and "+y+" Engineering");
```

➤ **Comparison Operators** (used in logical statments)

Given that **x=5**, the table below explains the comparison operators:

Operator	Description	Example
==	is equal to	x==8 is false
===	is exactly equal to (value and type)	x===5 is true x=== "5" is false
!=	is not equal	x!=8 is true
>	is greater than	x>8 is false
<	is less than	x<8 is true
>=	is greater than or equal to	x>=8 is false
<=	is less than or equal to	x<=8 is true

➤ **Logical Operators**

Given that **x=6 and y=3**, the table below explains the logical operators:

Operator	Description	Example
&&	and	(x < 10 && y > 1) is true
	or	(x==5 y==5) is false
!	not	!(x==y) is true

➤ **Conditional Operator**

Syntax

```
variablename=(condition)?value1:value2
```

Example

```
// x variable can have values > 0 or <0  
  
x = -2;  
  
// y will be true if x > 0 and will be false if x < 0  
  
y = (x>0)?true:false;  
  
switch(y)  
{  
    case true:  
        document.write("x is greater than zeo"); break;  
  
    case false:  
        document.write("x is lower than zeo"); break;  
  
}
```

If Statement

```
if (condition1)
{
  // code to be executed if condition1 is true
}
else if (condition2)
{
  // code to be executed if condition2 is true
}
else
{
  //code to be executed if condition1 and condition2 are not true
}
```

Example:

```
<script type="text/javascript">
var d = new Date()
var time = d.getHours()
if (time<10)
{
  document.write("<b>Good morning</b>");
}
else if (time>10 && time<16)
{
  document.write("<b>Good day</b>");
}
else
{
  document.write("<b>Hello World!</b>");
}
</script>
```

Switch

perform different actions based on different conditions

Syntax

```
switch(n)
{
  case 1:
    execute code block 1
    break;
  case 2:
    execute code block 2
    break;
  default:
    code to be executed if n is different from case 1 and 2
}
```

Example

```
x = -1;

switch((x>0))
{
  case true:
    document.write("True : ");
    break;
  case false:
    document.write("Flase : ( ");
    break;
  default:
    document.write("Zeroooooo");
}
```

Loop

execute a block of code a specified number of times or while a specified condition is true.

For Loop

```
for (var=startvalue;var<=endvalue;var=var+increment)
{
  code to be executed
}
```

While Loop

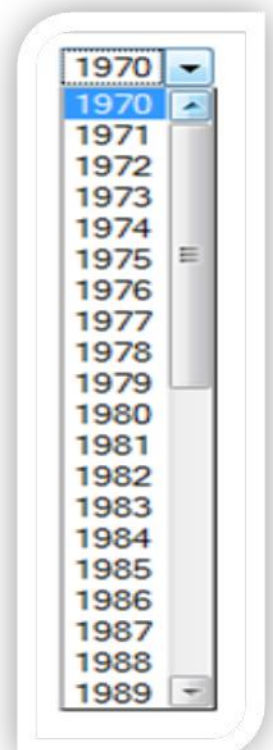
```
while (var<=endvalue)
{
  code to be executed
}
```

Do ... While

```
do
{
  code to be executed
}
while (var<=endvalue);
```

Example:

```
1 <html>
2   <head>
3     <title>Learn JavaScript</title>
4   </head>
5 <body>
6 <select name="age" >
7   <script type="text/javascript" >
8     for(i=1970;i<=2009;i++)
9     {
10      document.write("<option value='"+i+"' >"+i+"</option>");
11    }
12  </script>
13 </select>
14 </body>
15 </html>
```



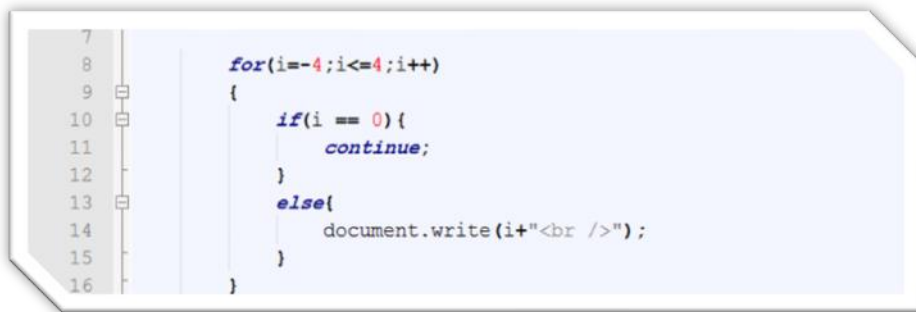
Break:

break the loop and continue excuting the code

Continue:

break the current loop and continue to next loop

Example



For ... in

loop throught elements of an array or properties of an object

Syntax

```
for (variable in object)
{
    code to be executed
}
```

Example

```
var x;
var mycars = new Array();
mycars[0] = "Saab";
mycars[1] = "Volvo";
mycars[2] = "BMW";

for (x in mycars)
{
    document.write(mycars[x] + "<br />");
}
```

Fuctions:

- function is excuted when called or when an event occurred.
- To keep the browser from excuting the script on load, you can put your code in a funtion and

Syntax

```
function functionname(var1,var2,...,varX)
{
  some code
}
```

- Variables in function can be accessed only be function.

Examples

```
1 <html>
2   <head>
3     <title>Learn JavaScript</title>
4     <script type="text/javascript" >
5       function show_msg()
6       {
7         alert("Hello World !");
8       }
9     </script>
10  </head>
11  <body>
12    <input type="button" value="ClickMe" onclick="show_msg()" />
13  </body>
14 </html>
```

```
1 <html>
2   <head>
3     <title>Learn JavaScript</title>
4     <script type="text/javascript" >
5       function add(x,y)
6       {
7         z = x + y ;
8         return z;
9       }
10    </script>
11  </head>
12  <body>
13    <input type="button" value="ClickMe" onclick="alert(add(3,4))" />
14  </body>
15 </html>
```

```
1 <html>
2   <head>
3     <title>Learn JavaScript</title>
4     <script type="text/javascript">
5       function product(a,b)
6       {
7         return a*b;
8       }
9     </script>
10  </head>
11  <body>
12    <script type="text/javascript">
13      document.write(product(4,3));
14    </script>
15  </html>
```

PopUp Boxes

1) Alert Box

make sure that the information comes from the user

```
<input type="button" value="ClickMe" onclick="alert('Hello! \n I am in new line :) ');" />
```

2) Confirm Box (Ok, Cancel)

if we want user to varify or accept something

```
1 <html>
2   <head>
3     <title>Learn JavaScript</title>
4     <script type="text/javascript">
5       function show1()
6       {
7         r = confirm("Are you sure ?");
8         if(r == true){
9           document.write("you clicked Ok !");
10        }else{
11          document.write("you clicked Cancel !");
12        }
13      }
14    </script>
15  </head>
16  <body>
17
18    <input type="button" value="ClickMe" onclick="show1()" />
19
20  </html>
```

3) Prompt Box

when we want user to input some thing before entering the page

```
1 </html>
2 <head>
3   <title>Learn JavaScript</title>
4   <script type="text/javascript">
5     function show1()
6     {
7       n = prompt("Enter your name:", "default name");
8       if(n != ""){
9         document.write("Your name is "+n);
10      }
11    }
12  </script>
13 </head>
14 <body>
15
16   <input type="button" value="ClickMe" onclick="show1()" />
17
18 </body>
</html>
```

Special Text

You can use backslash sign (/) to add special characters to the text

Code	Outputs
\'	single quote
\"	double quote
\&	ampersand
\\	backslash
\n	new line
\r	carriage return
\t	tab
\b	backspace
\f	form feed

Examples:

```
\\ new line

alert("Line one \n Line2 ");
```

```
\\ writing double quote inside another

<input type="text" onclick='alert("This is \"JavaScript\"")' />
```

```
\\ tab

<input type="text" onclick='alert("\t Hello \n new line")' />
```

Advanced Topics

Read & study advanced JS topics in

<http://www.w3schools.com/js/default.asp>

Objects, Strings, Date, Array, Math, Browsers, Cookie

- Js Events http://www.w3schools.com/js/js_events.asp
- Js Examples http://www.w3schools.com/js/js_examples.asp
- Js Examples2 http://www.w3schools.com/js/js_examples_2.asp
- Js Examples3 http://www.w3schools.com/js/js_examples_3.asp
- Lean HTML DOM <http://www.w3schools.com/html/dom/default.asp>

Here You can fine Many Js Codes:

- <http://www.star28.com/java/index.html>
- http://www.w3schools.com/html/dom/dom_examples.asp
- <http://www.scriptsearch.com/JavaScript/>
- <http://www.hotscripts.com/category/javascript/>
- <http://www.dhtmlgoodies.com/>
- <http://www.freecodemomma.com/links.php?cid=5>
- <http://www.webscriptsdirectory.com/JavaScript/>
- <http://www.webmonkey.com/codelibrary/tag/javascript>

Resources:

www.w3schools.com

www.google.com

www.eng-mmfm.com

Don't forget :

- **Great minds discuss ideas; Average minds discuss events; Small minds discuss people.**
- **Life Gives answers in 3 ways..It says yes and gives you what you want, it says no and gives you something better,it says wait and gives you the best**
- **You do not love people because they look beautiful! they look beautiful because you love them**
- **Happiness is not a destination....it's a method of life**

- **Yesterday is history. Tomorrow is mystery. Today is a gift that is why it is called "THE PRESENT".**
- **a bad wrok man ,blames his tools**
- **DON'T cry over anyone who won't cry over you**
- **Some people make the world SPECIAL just by being in it**
- **Change before you have to.**

Good Luck